WORKED SOLUTION

STEP II Specimen paper, Question 7

Pure • Integrals of Step Functions

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q7

The key idea

The integrand $[\mathrm{e}^x]$ is a step function: it is constant on each interval where e^x stays between two consecutive integers. So we find those intervals, integrate a constant over each one, and add up. The sum that comes out looks awkward, but most of its logarithms cancel in pairs once we line up equal terms.

Solution

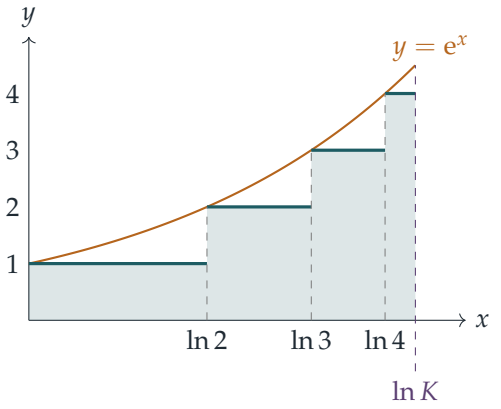
Where the integrand is constant

We take $K \geq 1$, so that $\ln K \geq 0$ and $N = [K] \geq 1$. (If $0 < K < 1$, then for x between $\ln K$ and 0 we have $0 < \mathrm{e}^x \leq 1$, and $[\mathrm{e}^x] = 0$ except at the single point $x = 0$, so $I = 0$; also $N = 0$ and $N \ln K - \ln 0! = 0$. The result holds there too.)

For an integer $k \geq 1$,

$$k \leq \mathrm{e}^x < k+1 \iff \ln k \leq x < \ln(k+1),$$

because ln is increasing. So $[\mathrm{e}^x] = k$ on $[\ln k, \ln(k+1))$. On the range of integration $0 \leq x \leq \ln K$ we have $1 \leq \mathrm{e}^x \leq K$, so $[\mathrm{e}^x]$ takes the values $1, 2, \ldots, N$, and the last value N holds on $[\ln N, \ln K]$, since $N \leq \mathrm{e}^x \leq K < N+1$ there.



The shaded area is I (drawn with $K = 4.5$, so $N = 4$).

Adding up the pieces

Integrating the constant on each interval,

$$I = \sum_{k=1}^{N-1} k(\ln(k+1) - \ln k) + N(\ln K - \ln N).$$

(When $N = 1$ the sum is empty; when $K = N$ the last term is zero. Neither case needs separate treatment.)

Ask yourself How do we simplify $\sum k(\ln(k+1) - \ln k)$? It doesn't telescope as it stands.

Split it into two sums and shift the index in the first, putting $j = k + 1$:

$$\sum_{k=1}^{N-1} k \ln(k+1) = \sum_{j=2}^{N} (j-1) \ln j.$$

Now both sums are sums of multiples of $\ln j$, and we can collect the coefficient of each $\ln j$. Using $\ln 1 = 0$,

$$\begin{aligned}\sum_{k=1}^{N-1} k(\ln(k+1) - \ln k) &= \sum_{j=2}^{N} (j-1) \ln j - \sum_{j=2}^{N-1} j \ln j \\ &= (N-1) \ln N + \sum_{j=2}^{N-1} ((j-1) - j) \ln j \\ &= (N-1) \ln N - \sum_{j=2}^{N-1} \ln j.\end{aligned}$$

Adding the final piece $N \ln K - N \ln N$:

$$\begin{aligned} I &= N \ln K - \ln N - \sum_{j=2}^{N-1} \ln j \\ &= N \ln K - \sum_{j=1}^{N} \ln j \\ &= N \ln K - \ln(N!), \end{aligned}$$

as required.

A slicker view

Instead of slicing the shaded region into vertical strips, slice it into horizontal layers. For $1 \leq k \leq N$, the layer between heights $k-1$ and k is present exactly where $[\mathrm{e}^x] \geq k$, that is, where $\mathrm{e}^x \geq k$, i.e. for $\ln k \leq x \leq \ln K$. That layer has area $\ln K - \ln k$, so

$$I = \sum_{k=1}^{N} (\ln K - \ln k) = N \ln K - \ln(N!)$$

at once. Formally this uses $[y] = \sum_{k=1}^{N} \mathbf{1}[y \geq k]$ for $1 \leq y < N+1$, and then integrates term by term. It is the same rearrangement as the index shift above, done geometrically.

Checks

- $K = 1$: the range of integration has length zero and the formula gives $1 \cdot 0 - \ln 1 = 0$.
- $K = \mathrm{e}$: here $N = 2$, and directly $I = \ln 2 + 2(1 - \ln 2) = 2 - \ln 2 \approx 1.3069$, matching $2 \ln \mathrm{e} - \ln 2$.
- For $K = 10$ the formula gives $10 \ln 10 - \ln(10!) \approx 7.9214$, and midpoint quadrature of $[\mathrm{e}^x]$ over $[0, \ln 10]$ agrees to 5×10^{-6}. Random values of K in $[1, 40]$ also agree.

Common slips

- Running the sum up to N with the interval $[\ln N, \ln(N+1)]$. The top of the range is $\ln K$, not $\ln(N+1)$, so the last piece must be handled separately.
- Getting the end terms wrong when re-indexing: write out the sums for $N = 3$ to check.

Beyond STEP

Swapping $\sum k(a_{k+1} - a_k)$ for $-\sum a_k$ plus boundary terms is *summation by parts* (Abel summation), the discrete analogue of integration by parts. Taking $K = N$ and using $\mathrm{e}^x - 1 < [\mathrm{e}^x] \leq \mathrm{e}^x$ gives $N \ln N - N + 1 \leq \ln(N!) \leq N \ln N - N + 1 + \ln N$, a first step

towards Stirling's approximation. The same idea, written as a Riemann–Stieltjes integral $\int f \, \mathrm{d}[t]$, turns sums into integrals throughout analytic number theory; see Apostol, *Introduction to Analytic Number Theory*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.